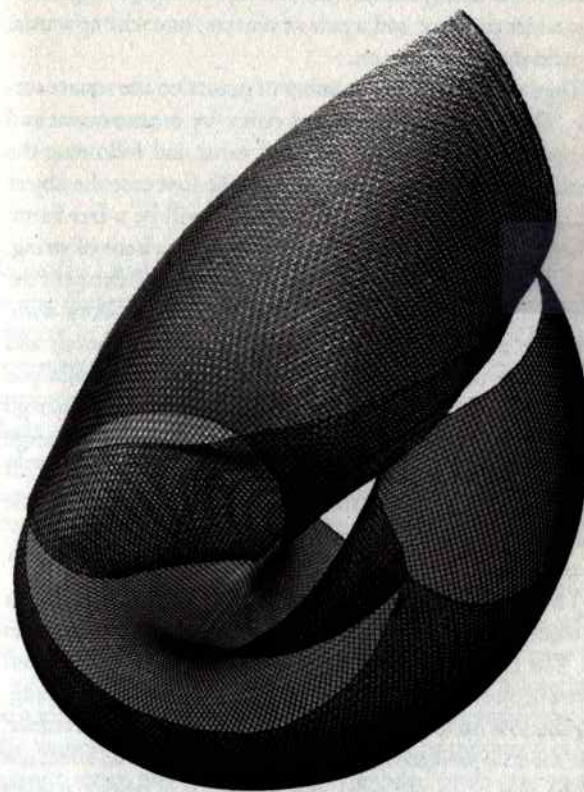


Concave-Convex Forms



These plastic forms are made out of a square of metal netting with a mesh of about four or five millimetres. The square itself can be of any size, and depends on the width of the netting you buy. When you have cut the square, you need to

turn the wires on the cut side back on themselves, so as to prevent the last wire (parallel with the cut) from shredding loose. This needs patience and a pair of pincers: one wire upwards, the next down, and so on.

Then you decide on a number of points on the square surface. These may be determined either by measurement and proportion or freely, by trial and error and following the dictates of the form as it emerges. In the first case the object will be mathematical, in the second it will be a free form. When you have marked these points with a knot of string, begin bending one of the corners of the net towards one of the points. The net is flexible and not difficult to work with. Fold the net slowly back on itself, keeping it smooth and round with the hand, until the corner touches the point you are aiming for. Tie it there with a piece of fine wire. Then go to work on the other corners, and a plastic form will emerge from a piece of netting that was perfectly flat before. This form will be semi-transparent and will have the same plastic characteristics as a shell, or as certain topological or mathematical forms.

I began making these forms in 1948, and exhibited them as objects to be hung from the ceilings of restaurants and so on. The slightest movement of the air makes them turn, and if you project light on to them they throw a constantly changing shadow on to the walls and ceiling, or even into a corner. The shadow projected looks like a drawing, with an effect not unlike that of an old print, but when it is in motion it gives beautiful *moiré* effects that appear and dissolve like a cloud.

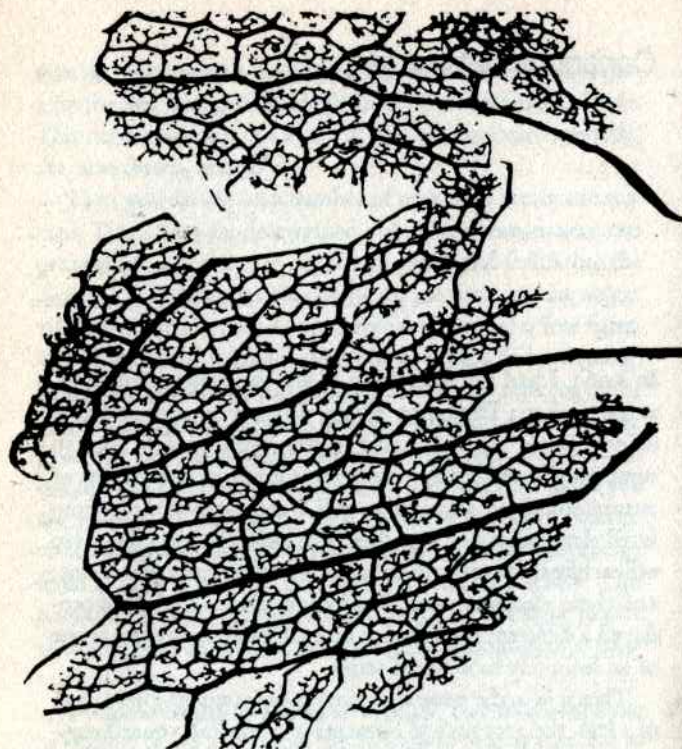
These objects were a by-product of my 'useless machines', and they too are fragile and therefore not easily saleable compared with things cast in bronze and destined to sit in museums, perfectly still, for centuries.

Continuous Structures

In 1961 I had the first show of my 'continuous structures' at the Galleria Danese in Milan. There were about a dozen objects in the show, ranging in size from one foot high to something over six feet. They were made of bent sheets of natural anodized aluminium, and were composed of modulated elements joined together. The basic elements, joined to others like them, gave these structures their particular forms; and these elements were of different kinds. Each kind produced a different final form, and each object emerged as one of an infinitely modulated series.

Thus it is in the natural structures of matter: the nervation of a leaf, the geometrical elements of a mineral, a quartz crystal or cubic pyrites. In the same way these simple elements confer form upon objects that can be thought of as belonging to a kind of man-made nature, structured in the same way as the natural things we know. They could be produced in an infinite series, just as theoretically a mineral could take on an infinite number of forms while at the same time remaining recognizably itself, classifiable according to its particular kind.

The elements of the continuous structures are based on the elementary formulae of the square and the right angle. The



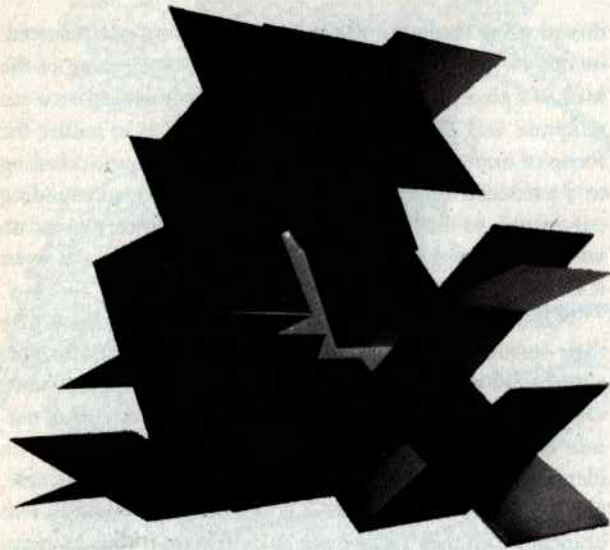
simplest of these elements consists of two squares soldered together along one side, and each having a straight cut made from the centre to the middle of another side, parallel to the joint. Two of these elements can be fitted together to form the cell of a certain kind of structure. This is the basic form. Other elements slotted into one another to form a chain, until the last slots into the first, make up a continuous structure that may be composed of any number of equal parts. In addition, the object may be put together in a number of ways, according to whether the pieces are mounted all facing in one

direction, or turned in the opposite direction, or alternated, or otherwise varied. The final result has something of the look of a piece of abstract sculpture, or of something between sculpture and the mineral world. And just as in nature the forms of minerals or plants or anything that grows according to a particular internal structure are limited by surrounding conditions, so the limits of a continuous structure depend on what the owner wants and the surroundings he intends to put it in.

Natural forms are continually modified during growth by their surroundings. Theoretically all the leaves of a single tree should be identical, but this could only happen if they were able to grow in surroundings completely devoid of outside influences and variations. All oranges should have an identical round shape. But in reality one grows in the shade, another in the sun, another in a narrow space between two branches, and they all turn out to be different. This diversity is a sign of life as it is actually lived. The internal structures adapt themselves and give birth to many diverse forms, all of the same family but different.

There are no works of art that show us this aspect of nature. In the usual way a piece of sculpture shows us one aspect of reality and one only. These continuous structures, with their capacity for variation according to the likings and moods of the spectator, bring to mind a whole realm of nature as yet unexplored by the plastic arts.

Naturally, this puts an end to the already tarnished image of the work of art as a rare and even unique thing, independent of what it expresses. Each continuous structure is reproduced in a certain number of 'dissimilar' copies, dissimilar in the number of elements used and in their arrangement. Two continuous structures made of the same number of



elements can be different, according to the temperaments of whoever assembled them.

In the Japanese magazine *Graphic Design* (No. 3, p. 40) there is a picture of one of these structures. Assembled by Shuzo Takigouchi, it is completely different from the one owned by Michel Seuphor of Paris, though it is made of the same number of identical basic elements.

Another thing about these structures is that unlike conventional pieces of sculpture they do not have a base, a top and a bottom, a back and a front. They can be placed any way up, or hung on a wire or a wall. Any similarity to abstract sculpture is purely accidental, in that certain abstract sculptures could perhaps catch one aspect of one phase of a continuous structure. The structures do not have a 'composition', an interplay of masses and spaces fixed once and for all. The

only fixed and invariable thing is the basic element, but even this becomes variable according to how many are used and how they are arranged in the object as a whole.

This too is a requirement of art today. We need to give the spectator more room to penetrate into the work itself, and works which allow this are called 'open'. It is a form of art that adapts itself to the artistic sense of the beholder. In times past people wanted the artist to explain in very clear terms exactly how he saw the world in every detail. The beholder was content to react to the personality of the artist, who in everyone's eyes became a genius, the greatest, the one whom nobody could imitate. Today the person who looks at a work of art is more sensitive, more accustomed to simultaneous and intense stimuli, to brand new technical and scientific concepts, so he is no longer so interested in a 'closed' work of art. Art that is too defined, conclusive, and limited to one aspect of a thing, leaves a man of today standing isolated and apart: either he accepts the *fait accompli* or he gets nothing from it. There is very little actual participation involved. Everything that does not coincide with the particular vision of the artist has to be excluded. But in an open work of art a person participates much more, to the extent of being able to change the work of art according to his state of mind.

The Tetracone

The tetracone is an object produced on a small scale for the purposes of programmed art. It consists of a cube with sides eight inches long, within which four cones revolve slowly at different speeds. The cube is open on two (opposite) sides, so that the cones inside it may be seen from front and back. They are arranged so as to fill the entire areas of the spaces thus presented to the view. From the front one sees one combination of colours, from the back the same combination but with the opposite colours.

Each cone contains a small motor and is fixed to the inside of one of the walls of the cube. The cones are cast in aluminium with an internal cavity that contains the motor. All the wires pass through grooves cut into the inside of the box, which is then covered with black plastic. On the base there is nothing but a plug for electric current and a switch.

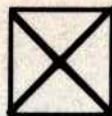
The dimensions of the box are the basis of all the other forms. Each element is related to every other one and to the whole, not according to the old rules of the Golden Section, which do not apply in this case, but by simple geometrical relationships. The diagonals of the sides of the box determine the size of the cones. The diameter of the base of each cone is equal to the internal length of the sides of the box,

while its height from base to apex is equal to half that amount. The surface of each cone, if projected on to a flat surface, is equal to three quarters of the circle which has a radius of half the diagonal of the side of the box. Each cone is painted in two complementary colours, half one colour and half the other. In this case the colours are red and green. Complementary colours produce an optical vibration at the dividing line between them, and this gets rid of the material. I had better explain this: if you look at complementary colours in the right lighting conditions you will lose all awareness of the material they are painted on. You will therefore be looking purely at the colour and not at the material.

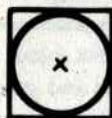
The programming part of the operation consists in deciding the speeds at which the four cones will turn, and the combination of the number of seconds that each cone takes to make a complete turn. In this way one can establish not only the three spatial dimensions but also an effective (as opposed to poetical or metaphysical) dimension of time, which in the tetracone I am describing here is 1,080 seconds. This means that theoretically the combinations of colour repeat themselves every eighteen minutes.

The tetracone therefore offers the viewer a changing combination of two complementary colours. Looking at the object, with its shifting chromatic effect, is supposed to lead the viewer to meditate on the mutability of nature (which in traditional art is presented as static). Because of the slowness with which the change occurs, the colour combinations are at first perceived one by one, like the single frames in a film, but if one watches for longer the effect becomes that of a continuous transformation, from all green to all red, in the course of eighteen minutes.

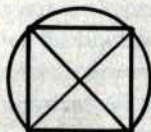
The art of the past (painting and sculpture) has accustomed



setting a position for cones in a cubic space



base of cone = side of the square



the flat surface of each cone takes up $\frac{3}{4}$ of a circle



the surface of each cone is divided in two equal parts of complementary colours



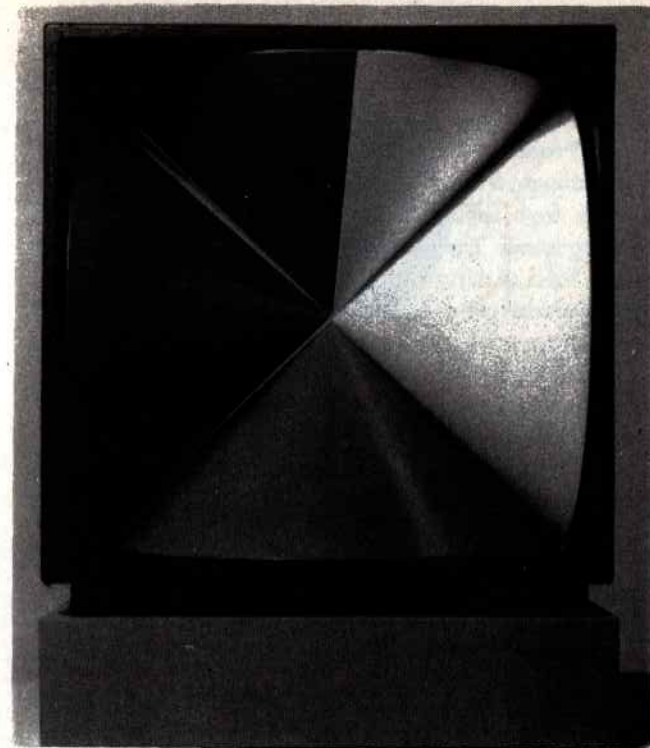
kinetic distribution of the speeds



time, in seconds, for each full turn



kinetic direction of four motors built into the cones



'As long as we try to slow up and limit spontaneous change by means of a static symbol we will never be able to understand or to act in a really effective way.' (Alexander Dormer)

'The greatest freedom comes from the greatest strictness.' (Paul Valéry)

us to seeing nature as static: a sunset, a face, an apple, all static. People go to nature looking for images such as these static things, whereas an apple is in fact a moment in the process from apple-seed to tree, blossom, fruit. In nature nothing is still. The idea of nature fixed at 1 June 1969 or a face fixed at thirty-two years and eight days old is completely unreal; quite apart from the fact that if we stop nature we shall never be able to understand her.

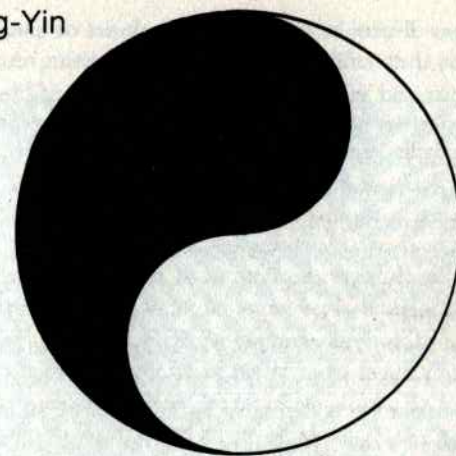
When we fly over the Pole we see a sunset that lasts for hours, and then slowly changes into dawn. Sunset and dawn are the same thing; day and night are both continuous in the world. They never stop. What kind of art can tell us about these facts of nature? What have we learnt from the art of the past about the cyclic movements of nature? Or about the transformations of forms and colours? Think of a tree when it is a seed, and then imagine it tall and green and flowering, laden with fruit. Think of it in autumn, and in winter. All this is nature and nature is all of this, not just one moment of it.

The programmed art of today aims to show forms while they are in the process of becoming, and for this reason it cannot use forms such as painting and sculpture use. On the contrary, its means must be dynamic, and it must be prepared to make full use of motors and other industrial materials.

What really counts is the information which a work of art can convey, and to get down to this we have to abandon all our preconceived notions and make a new object that will get its message across by using the tools of our own time.

'The principle of a form is not it is but one does.' (Bauhaus)

Yang-Yin



This Yang-Yin symbol is of Chinese origin, and is more than three thousand years old. It represents the unity created by a balance between two opposing forces that are equal and contrary.

This unity is visibly represented by a disc made up of two equal parts, one black and one white. These parts seem to be in constant movement in a clockwise direction.

The two opposing forces are interpreted as natural forces, and from their balance comes life itself. Yang is the positive, active, masculine force, and subsists in dryness, heat, hardness, the sky, light, the sun, fire. It is firmness and brightness. Yin is the negative feminine principle present in everything passive, such as cold, damp, softness, mystery. It is in all mysterious, secret, evanescent, cloudy things, and in everything inactive. The shadow on the north side of a hill, the estuary of a river, earth and water: these are Yin.

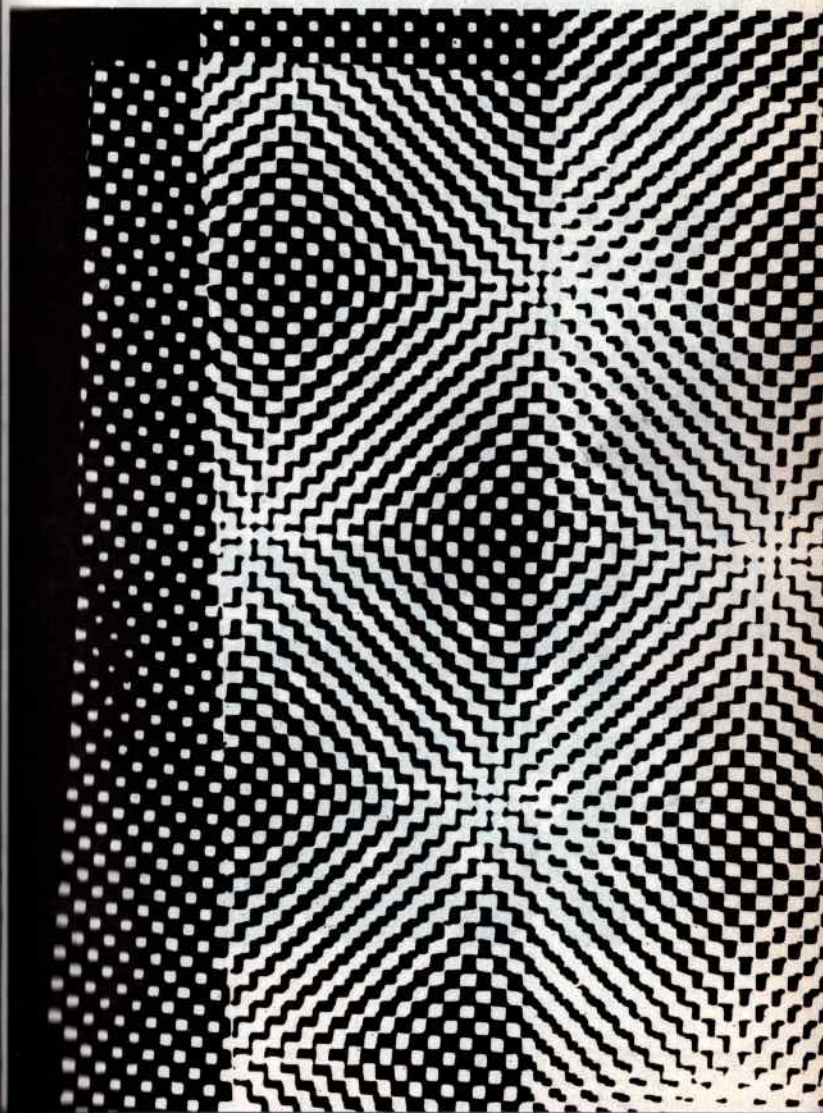
Millions of people, amazed by the glories of nature and stunned by the terrible havoc of natural calamities, recognize this definite and mobile symbol as the symbol of life itself.

We ourselves get tired, and rest in order to recover our equilibrium. If our eyes are struck by a sudden bright red light, we see a complementary green colour until our retina has returned to normal.

This rule about complementary colours, or the balance of opposites in general, must always be borne in mind in some kinds of work.

In his book *The Hidden Persuaders*, Vance Packard describes an experiment made by psychologists to determine the best colour for a detergent packet. Three trial packets were made up, one yellow, one blue, the third yellow and blue. Housewives chose the yellow and blue packet (complementary colours in equilibrium), saying that in their opinion the yellow packet contained a corrosive detergent, and the blue packet a detergent that was too mild. The yellow and blue packet contained a detergent that was just right. It goes without saying that the detergent in all three packets was one and the same product.

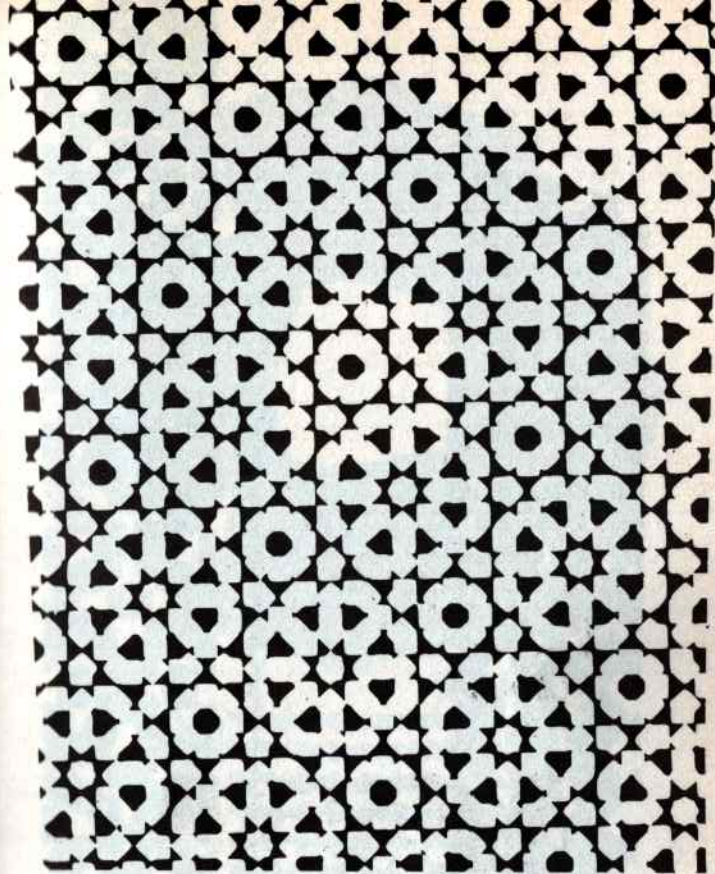
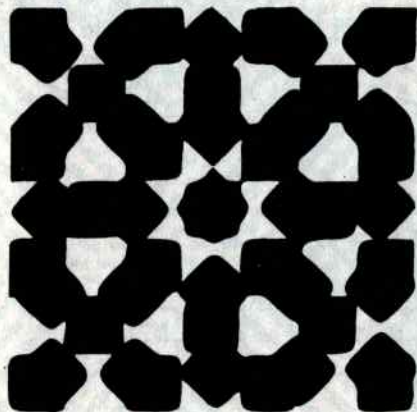
Moiré



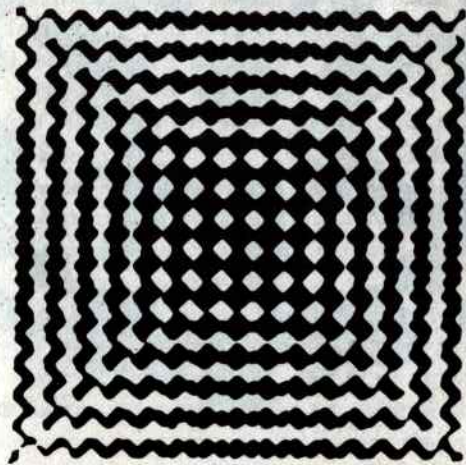
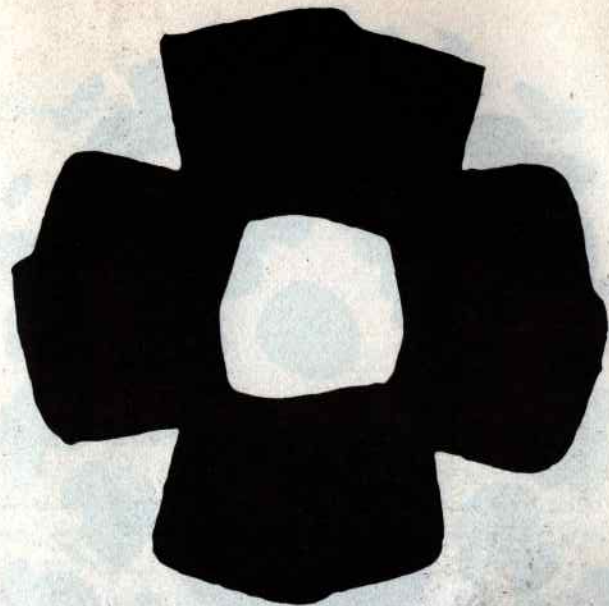
Printers use various kinds of screens. Some have round dots, others square, and the distance between the dots also varies. Take two screens with square dots which do not quite touch (not as on a chessboard, where the squares touch at the corners), printed in black on a sheet of transparent paper. Place one sheet on top of the other so that the patterns are superimposed and then shift the top sheet very slightly round. This will produce images that were not there before, symmetrical patterns like flowers, or like Arab decorative motifs.

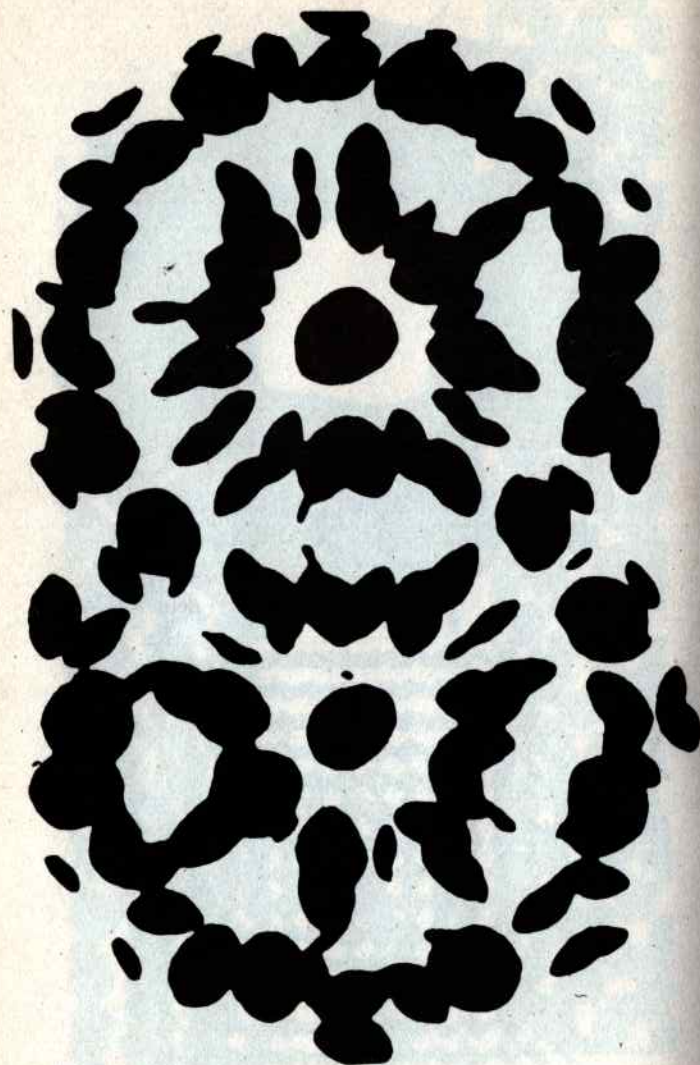
The images change according to the angle at which the sheets are placed to one another. Not only does one image change into another, but it is interesting to watch the actual process of transformation taking place.

My friend Marcello Piccardo and I once made a short film of this. We called it *Moiré*.



The equipment for filming and projection was set up on the basis of objective calculations, starting from basic elements and then developing according to the nature of the theme. The music was contrived by Pietro Grossi in his sound-research laboratory in Florence; it has the same structure as the images themselves. The film only lasts three minutes, because by then we had run through all possible variations of image.





Techné — art
Asobi — art, but also game (in ancient Japanese)

Direct Projections

Direct projections of what? Well, of a dragonfly's wing for example. Take the ordinary frame that you use for colour slides, get two squares of thin glass of the right size and sandwich the dragonfly's wing between them. Needless to say, you should take a wing from a dead dragonfly and not kill one specially to make an amusing experiment. You then project the wing with the ordinary projector that you use for photographs of your summer holidays, enlarged to six feet high for the greater delight of your guests. And *voilà*, you see everything. You see its design and how it is made. Then when you get bored with that (and I know just how quickly you do get bored) get a feather, just one of the tiny feathers from your grandmother's softest cushion. And there is the feather six feet long, absolutely beautiful, with its hundreds of filaments that you now notice for the first time are themselves long feathers.

If you can't find your grandmother's feather cushion, take a foam rubber one and cut a slice off it as thin as a Communion wafer (sorry Vicar). You will see the structure of the foam rubber, and it really is like a drawing of foam.

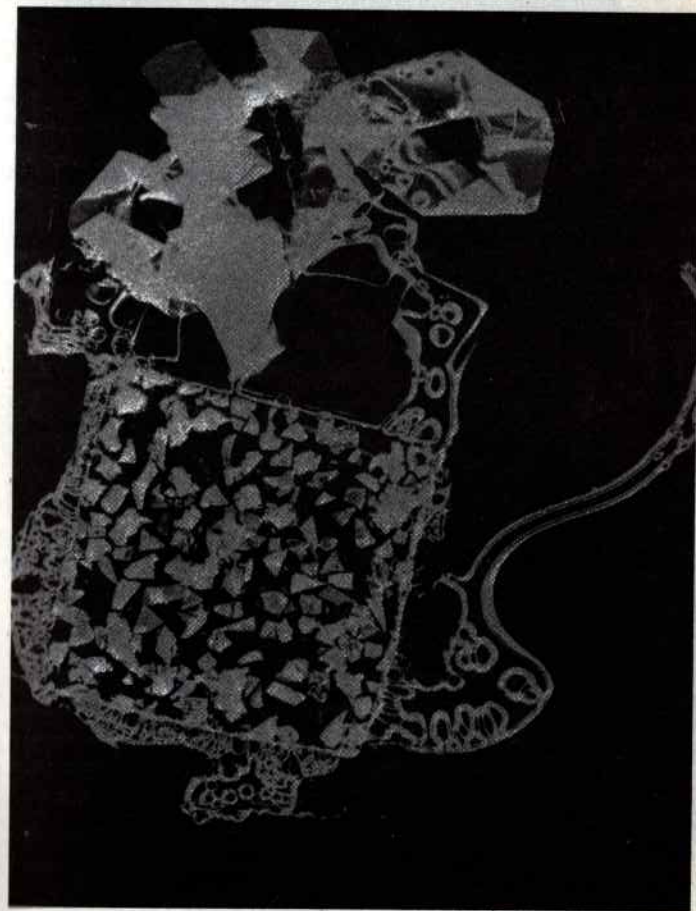
But seriously, how can we make full use of a slide-projector in this day and age?

We can use it as a means of expression, just like any other artistic technique. Just as one can buy tubes of paint for painting pictures, so one can obtain sheets of coloured Cellophane and other substances that can be used for projecting. They are very thin sheets and come in a wide range of colours, so that you have no lack of choice, from the lightest colours to the darkest. You can put two or three thicknesses of one colour to darken it, or you can superimpose different colours to get any mixture you like.

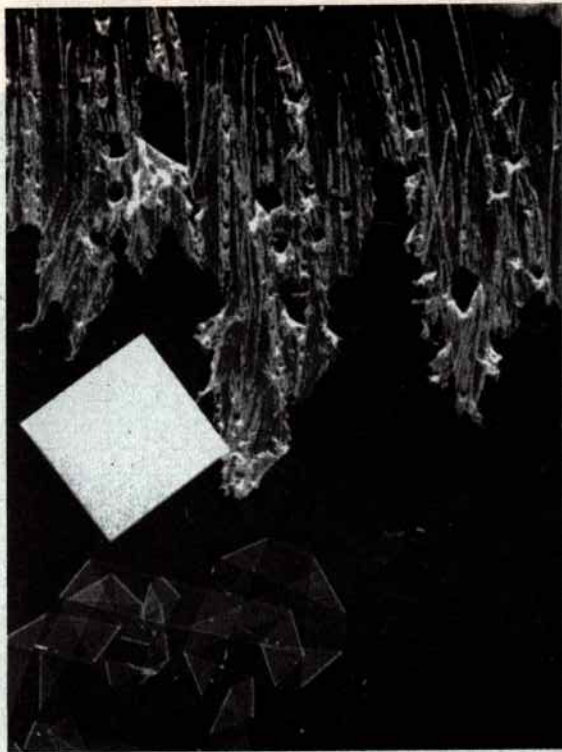
You can also find sheets of transparent coloured plastic, as thick as a visiting card. These can be scratched, engraved, burnt or dipped in solvents to obtain many different effects.

Apart from these there are materials which have a particular structure and therefore (when projected) a particular design, such as the nervation of leaves, some nylons and other artificial fabrics, mica, threads as fine as hairs, crystalline powders, and so on.

All these things should be used with the utmost freedom. It is no good trying to make a portrait of Aunt Mary or a view of Loch Lomond. At least, one *could* do it, but it would be a lot of trouble to get a rather poor result. The best thing is to use small pieces of these materials and put them together as the mood takes you, without any pretence at creating a masterpiece, working by trial and error. Simply by trying things out you will become involved in an absorbing game (the ancient Japanese word for art was *asobi*, which also means game). You will find that this game gives the chance of self expression, that certain colours and forms which come into being by chance provoke certain sensations, and that certain strange colours remind you of something. You will find out that these images evoke feelings and memories from the distant past: the colours of a terrace where one played



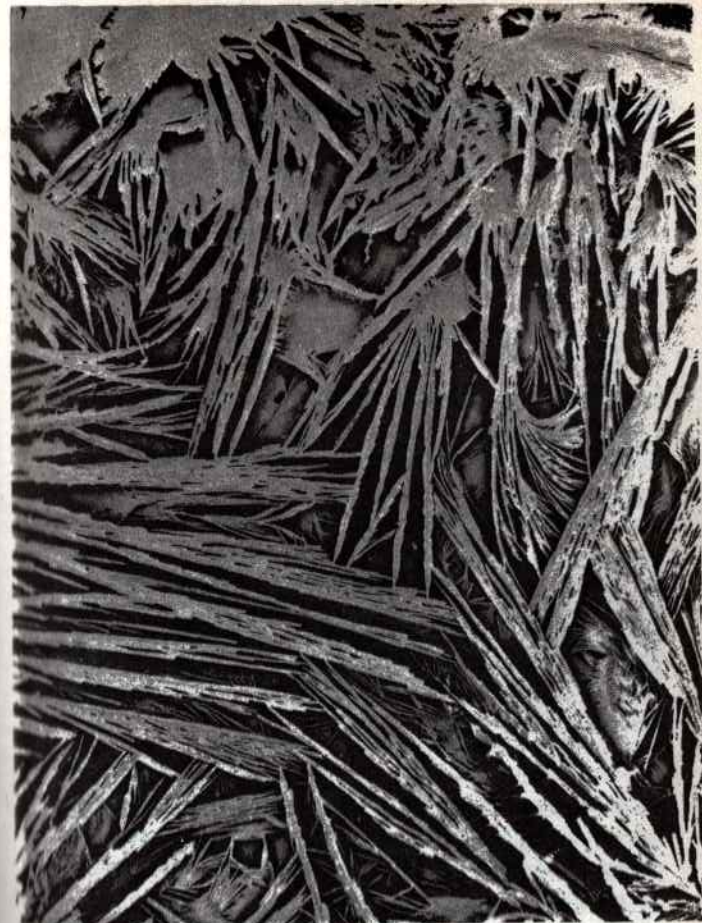
'To understand means to be capable of doing.' (Goethe)



as a child, the waters of a river full of rapids and whirlpools, a square in the shadow of a great looming castle (I mention things at random). Then you can ask the other viewers to talk about their reactions. It will be an interesting experiment.

When the experiment is over you will still have a collection of little slides to keep in a cigar box (one of those wonderful cigar boxes that smell of wood and good tobacco).

'When man ceases to create he ceases to live.' (*Lewis Mumford*)



Hyposulphite crystals in a lantern slide. The slide is used just like a normal negative, put in the enlarger and printed on normal paper.